

Chapter 1: Principles of Stratified Random Sampling

Stratified sampling is a probability sampling method where the target population is partitioned into non-overlapping, mutually exclusive, and collectively exhaustive subpopulations called strata. Within each stratum, independent random samples are selected using simple random sampling (SRS).

Key Advantages:

1. Precision: Reduces sampling error compared to unstratified simple random sampling.
2. Administrative convenience: Enables regional office field operations to work independently.
3. Guaranteed representation: Ensures minority or crucial sub-demographics are adequately surveyed.

Chapter 2: Allocation Techniques in Stratified Surveys

Once strata are established, sample size n must be allocated across strata ($h = 1, \dots, H$).

Common allocation strategies include:

- Proportional Allocation: Sample size in stratum h is proportional to stratum size N_h .
- Neyman Optimal Allocation: Minimizes variance for a fixed sample size when stratum variances differ.
- Optimum Cost Allocation: Minimizes survey costs subject to specified precision tolerances.